



Remote Access to Raspberry Pi

There are several ways to remotely access Raspberry Pi. Here we discuss the simplest among them, using secure shell (SSH) and virtual network connection, respectively

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Raspberry Pi is an excellent platform to learn programming and interfacing of a computer with the real world. But it is always annoying to move to a completely different setup to design applications around Raspberry Pi. Also, the handling of additional stuff like monitor, keyboard, powered USB hub and mouse all over the place is also a terrible pain. Soon you will get tired of all these.

The solution to this problem is hidden in the word 'remote access.' Remote access to Raspberry Pi can really help save you from all the clutter. You can do all sort of interesting stuff using your very own existing desktop without the need of connecting anything other than the power adaptor and Ethernet cable/Wi-Fi adaptor to Raspberry Pi. You will be able to remotely access Raspberry Pi over the local network.

There are several ways of doing this. Here we discuss the simplest among them, which uses secure shell (SSH) and virtual network connection, respectively.

Basic setup required

1. In Raspberry Pi, insert the SD card with a good image of the operating system (refer 'Getting Started With Raspberry Pi' in EFY April 2013 issue).
2. Power it up.
3. Set up the network on Raspberry Pi (refer 'Set Up Network for Raspberry Pi' in EFY May 2013 issue).

The network should be up and running before you configure Raspberry Pi for remote access.

Remote access to Raspberry Pi using SSH

SSH is nothing but a network protocol for secure data communication be-

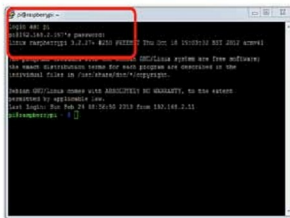


Fig. 1: An example of remote connection to Raspberry Pi using SSH. The red box shows the login details and IP address of Raspberry Pi

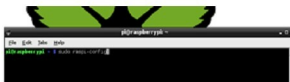


Fig. 2: raspiconfig command

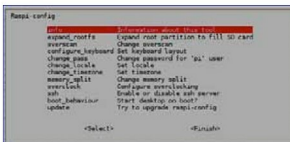


Fig. 3: raspiconfig utility



Fig. 4: Enable SSH server

tween two networked computers that are connected via a secure channel over an unsecured network. Here we access Raspberry Pi from a desktop computer over a local network using SSH.

With the help of SSH, you can effectively open a terminal session on Raspberry Pi from a host computer.

The first thing that you have to do is to enable your Raspberry Pi for using SSH. You can easily do this when you boot Raspberry Pi for the first time through 'raspi-config' utility. This utility appears automatically when you boot Raspberry Pi for the first time. If you missed to do this configuration setting during the first boot, don't worry, you can still do it easily by following these steps:

1. Open LX terminal on Raspberry Pi and enter the following command (refer Fig. 2):
`sudo raspi-config`
2. Now press 'Enter.' You will immediately see a window for raspiconfig utility (refer Fig. 3) from where you can do the configuration for SSH.
3. To enable the SSH server, scroll down to 'ssh' and hit 'Enter.' You